

AGORA

Post-Quantum Security Audit Pack

Program: /home/king_d/solana/testing/whirlpools

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Tool: Pinpoint SPECTRE 1.2.2

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Executive Summary

Total findings: 88

Cryptographic components inventoried (CBOM 1.6): 61

Findings by rule: CA-001=0, CA-002=50, CA-003=5, CA-004=0

CA-005: not applicable. PDAs are quantum-safe by construction.

Findings

CA-002

programs/whirlpool/src/instructions/adaptive_fee/initialize_adaptive_fee_tier.rs:24 | se
programs/whirlpool/src/instructions/adaptive_fee/initialize_pool_with_adaptive_fee.rs:32
programs/whirlpool/src/instructions/adaptive_fee/set_adaptive_fee_constants.rs:19 | sev=
programs/whirlpool/src/instructions/adaptive_fee/set_default_base_fee_rate.rs:13 | sev=M
programs/whirlpool/src/instructions/adaptive_fee/set_delegated_fee_authority.rs:13 | sev
programs/whirlpool/src/instructions/adaptive_fee/set_fee_rate_by_delegated_fee_authority
programs/whirlpool/src/instructions/adaptive_fee/set_initialize_pool_authority.rs:13 | s
programs/whirlpool/src/instructions/adaptive_fee/set_preset_adaptive_fee_constants.rs:13
programs/whirlpool/src/instructions/close_bundled_position.rs:31 | sev=Medium | Single-s
programs/whirlpool/src/instructions/close_position.rs:10 | sev=Medium | Single-signer au
programs/whirlpool/src/instructions/close_position_with_token_extensions.rs:14 | sev=Med
programs/whirlpool/src/instructions/collect_fees.rs:14 | sev=Medium | Single-signer auth
programs/whirlpool/src/instructions/collect_protocol_fees.rs:13 | sev=Medium | Single-si
programs/whirlpool/src/instructions/collect_reward.rs:15 | sev=Medium | Single-signer au
programs/whirlpool/src/instructions/delete_position_bundle.rs:23 | sev=Medium | Single-s
programs/whirlpool/src/instructions/increase_liquidity.rs:25 | sev=Medium | Single-signe
programs/whirlpool/src/instructions/initialize_fee_tier.rs:21 | sev=Medium | Single-sign
programs/whirlpool/src/instructions/initialize_reward.rs:9 | sev=Medium | Single-signer
programs/whirlpool/src/instructions/lock_position.rs:16 | sev=Medium | Single-signer aut
programs/whirlpool/src/instructions/open_bundled_position.rs:36 | sev=Medium | Single-si
programs/whirlpool/src/instructions/reset_position_range.rs:16 | sev=Medium | Single-sig
programs/whirlpool/src/instructions/set_collect_protocol_fees_authority.rs:11 | sev=Medi
programs/whirlpool/src/instructions/set_config_feature_flag.rs:12 | sev=Medium | Single-
programs/whirlpool/src/instructions/set_default_fee_rate.rs:13 | sev=Medium | Single-sig
programs/whirlpool/src/instructions/set_default_protocol_fee_rate.rs:11 | sev=Medium | S
programs/whirlpool/src/instructions/set_fee_authority.rs:11 | sev=Medium | Single-signer
programs/whirlpool/src/instructions/set_fee_rate.rs:13 | sev=Medium | Single-signer auth
programs/whirlpool/src/instructions/set_protocol_fee_rate.rs:13 | sev=Medium | Single-si
programs/whirlpool/src/instructions/set_reward_authority.rs:11 | sev=Medium | Single-sig
programs/whirlpool/src/instructions/set_reward_authority_by_super_authority.rs:14 | sev=
programs/whirlpool/src/instructions/set_reward_emissions.rs:19 | sev=Medium | Single-sig
programs/whirlpool/src/instructions/set_reward_emissions_super_authority.rs:11 | sev=Med
programs/whirlpool/src/instructions/swap.rs:17 | sev=Medium | Single-signer authority `t
programs/whirlpool/src/instructions/transfer_locked_position.rs:17 | sev=Medium | Single
programs/whirlpool/src/instructions/two_hop_swap.rs:17 | sev=Medium | Single-signer auth
programs/whirlpool/src/instructions/v2/collect_fees.rs:16 | sev=Medium | Single-signer a
programs/whirlpool/src/instructions/v2/collect_protocol_fees.rs:15 | sev=Medium | Single
programs/whirlpool/src/instructions/v2/collect_reward.rs:17 | sev=Medium | Single-signer
programs/whirlpool/src/instructions/v2/delete_token_badge.rs:13 | sev=Medium | Single-si
programs/whirlpool/src/instructions/v2/increase_liquidity.rs:33 | sev=Medium | Single-si

(continued on next page; pagination extended in task 13)

NIST IR 8547 Traceability Matrix

Maps each audit-pack section to a NIST IR 8547 post-quantum transition phase.

Phase	Rule(s)	Finding Count
Inventory	CA-001	0
Risk Prioritisation	CA-002, CA-003, CA-004	55
Hybrid Deployment	(no rules at this release)	n/a
Full Migration	(no rules at this release)	n/a

Hybrid Deployment and Full Migration are documented for framework completeness.

No CA rule fires for those phases at this release. Coverage expands in Tier 2.

Methodology

Pre-pass: Anchor expand subprocess invoked before tree-sitter walks source.

Expanded output is cached per program hash to avoid redundant re-expansion.

OtterSec downstream framing: CA-003 cross-program assumption findings surface migration cascade complexity following OtterSec's CPI-trust analysis framework.

CA-005 dropped: PDAs are quantum-safe by construction. Off-curve points derived via SHA-256 have no Ed25519 private key, so the Grover speedup falls beyond the attack horizon. Confirmed with Jump Crypto and Helius.

CA-004 is inventory and exposure only. It does not prescribe a hybrid Falcon co-signer. Protocol-layer migration (address format, ZK proof of Ed25519 seed) is owned by Anza and Firedancer, not by individual program developers.

Rule pack: pinpoint-rules-pq v0.1.0. CBOM format: CycloneDX 1.6.

PDF renderer: printpdf (pure Rust, no system library dependency).